

# ASM-I: Week 01 Notes

## Non-Parametric Methods

### Introduction to Hypothesis Tests

Non-Parametric (Distribution-free methods for testing hypothesis)

### Recap: Parametric Methods

#### 1. One Sample Mean

- Assumed Model:  $E[X_i] = \mu$
- Hypothesis Tests:
  - $H_0 : \mu = \mu_0$
  - $H_a : \mu \neq \mu_0$

- Test Statistic:

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$$

- Assumptions:

- $X_1, \dots, X_n \stackrel{iid}{\sim} F$
- If  $n$  is large,  $F$  can be any distribution (CLT applies).
- If  $n$  is small, we assume  $F \sim N$  (Normal Distribution).
- What if  $n$  is small and  $F$  is not Normal?

#### 2. Two Sample Mean

- Samples:  $X_1, \dots, X_n \sim F$  and  $Y_1, \dots, Y_m \sim G$
- Assumption:  $E[X] = \mu_1, E[Y] = \mu_2$
- Hypothesis:
  - $H_0 : \mu_1 = \mu_2$
  - $H_a : \mu_1 \neq \mu_2$
- Conditions:

- If  $F$  and  $G$  are both Normal  $\rightarrow$  **2-sample t-test**.
- If both  $m$  and  $n$  are large  $\rightarrow$  **Normal test**.

### 3. Variance Test (One Sample)

- Sample:  $X_1, \dots, X_n \sim F$
- Variance:  $\text{Var}(X_i) = \sigma^2$
- Statistic:  $\frac{(n-1)S^2}{\sigma_0^2} \sim \chi_{n-1}^2$
- $S^2$  is unbiased sample variance
- Hypothesis:  $H_a : \sigma^2 \neq \sigma_0^2$
- **Assumption:**  $F$  is Normal.

## Comparing Parametric vs. Non-Parametric

### 4. Two Sample Variance

- Statistic:  $\frac{S_1^2}{S_2^2}$
- Hypothesis:  $H_0 : \sigma_1^2 = \sigma_2^2$
- Test: **F-test**
- Assumption:  $F = N = G$  (Both distributions are Normal).

### 5. ANOVA (One-way)

- Equality of means of several samples.
- Valid under Normal assumptions.

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## Motivation for Non-Parametric Tests

What if the sample size ( $n$ ) is small and the parent distribution is unknown?

- If the parent distribution is known but *not* Normal, we may be able to construct the distribution of the test statistic (e.g., if  $F \sim \text{Exp}(\lambda)$ ,  $\sum X_i \sim \text{Gamma}(n, \lambda)$ .  $F \sim \text{Pois}(\lambda)$ ,  $\sum X_i \sim \text{Pois}(n\lambda)$ ) Then we can construct the test statistic under  $H_0$  using the knowledge of probability theory.
- However, often we do not know the distribution.

**Definition:** The tests we shall construct are **Distribution Free**.

- We do not assume any parametric model (a model with known form but finite number of unknown parameters).

### Why not always use Non-Parametric methods?

- The flexibility is attractive but comes with a cost.

- When a parametric model is true/correct, then using that parametric test is “better”.
- **Power:** For hypothesis testing, both parametric and non-parametric tests will have the same significance level  $\alpha$ . But the parametric test (under the correct parent model) will have **more power**.
- **Confidence Intervals:** Both procedures will give the same average probability (coverage), but the parametric interval will be **shorter/narrower** (more precise).

## The Sign Test (One Sample Median Test)

Use non-parametric techniques if the distribution of the data is unknown.

### One Sample Median Test

(Alternative to one sample t-test)

#### Concept:

- Generally for  $F$ , mean may not exist (e.g., Cauchy), but **Median always exists**.
- Median equals Mean if  $F \approx N$  (Normal).
- Definition: Median  $M = \min\{a : F(a) \geq 1/2\}$ .

#### Example Scenario: Arsenic Concentration

- Sources statlib (CMU) Arsenic in drinking water
- Data:  $X_1, \dots, X_{21}$  (Arsenic in toenail clippings of 21 individuals).
- Hypothesis:  $H_0 : \theta = 0.2$  vs  $H_a : \theta > 0.2$  (Right Tailed).
- Let Median( $X_i$ ) =  $\theta$ .

**Procedure:** 1. Under  $H_0$ , observe  $P(X_i > 0.2) = 1/2$ .

2. Sample  $X_1, \dots, X_n \stackrel{iid}{\sim} F$  (Independent).
3. Define Indicator Variable  $Y_i$ :

$$Y_i = \begin{cases} 1 & \text{if } X_i > \theta_0 \\ 0 & \text{if } X_i \leq \theta_0 \end{cases}$$

4. Test Statistic:

$$Y = \sum_{i=1}^n Y_i \sim \text{Bin}\left(n, \frac{1}{2}\right)$$

(Since  $X_i$  are independent,  $Y_i$  are independent Bernoulli trials).

#### Rejection Region:

- Reject for large values of  $Y$ .
- $P(Y > C) = \alpha$  (Right Tailed).
- Depending on the type of  $H_a$ , you may need a Left Tailed or Two Tailed rejection region.

# Sign Test Analysis & Wilcoxon Intro

**Randomness Check:** Randomness test is necessary.

**Example Calculation:**

- $H_a : \theta > 0.2$  and observed  $y = 10$  with  $n = 21$ .
- P-value calculation (using R): 2. `pbinom(10, 21, 0.5)=1`.
- If p-value  $> \alpha$ , do not reject  $H_0$ .

**Large Sample Approximation:**

- If  $n$  is large, use Normal approximation to Binomial (converges to normal quickly  $n \geq 10$ ) and not to general F (may take long time, much larger  $n$  needed).
- Test Statistic  $Y \sim \text{Bin}(n, 0.5)$ .
- $n = 21, y = 10$
- $Z = \frac{Y - np}{\sqrt{npq}} = \frac{Y - 0.5n}{0.5\sqrt{n}}$ .
- 2. `pnorm(Z)=0.8`
- Do not reject  $H_0$  at any  $\alpha < 0.8$ .

**Summary of Sign Test:**

- It is a one-sample median test looking at the sign of  $X_i - \theta_0$ .
  - Widely applicable. The only assumption is that  $F$  is **continuous**.
  - **Drawback:** Low power under many alternatives. It ignores the *magnitude* of the difference.
  - Example: A density could be symmetric but shaped such that signs are balanced even if the bulk of data is far off.
- 

## Wilcoxon Signed Rank Test

- Assumes  $F$  is **continuous** and **symmetric**.
- Ranks are distribution free.
- **Idea:** Give more weight to  $X_i$  that are much larger than  $\theta_0$  (large deviation).

**Procedure:**

1. Calculate absolute differences:  $|X_i - \theta_0|$ .
2. Assign Ranks  $R_i$  to these absolute differences.
3. Max Sum of Ranks:  $\sum R_i = \frac{n(n+1)}{2}$ .
4. Test Statistic:

$$T = \sum R_i Y_i$$

(Where  $Y_i$  is the indicator of positive sign, as defined before).

$$Y_i = \begin{cases} 1 & \text{if } X_i > \theta_0 \\ 0 & \text{if } X_i \leq \theta_0 \end{cases}$$

# Wilcoxon Signed Rank Example

## Example: Heart Rate

- $H_0$  : Median = 70 vs  $H_a$  : Median > 70.
- $\theta_0 = 70$ .
- Data ( $X_i$ ): 73, 82, 87, 68, 106, 60, 97.

## Calculation Table:

| $X_i$ | $X_i - \theta_0$ | Sign ( $Y_i$ ) | $ X_i - \theta_0 $ | Rank ( $R_i$ ) |
|-------|------------------|----------------|--------------------|----------------|
| 73    | +3               | 1              | 3                  | 2              |
| 82    | +12              | 1              | 12                 | 4              |
| 87    | +17              | 1              | 17                 | 5              |
| 68    | -2               | 0              | 2                  | 1              |
| 106   | +36              | 1              | 36                 | 7              |
| 60    | -10              | 0              | 10                 | 3              |
| 97    | +27              | 1              | 27                 | 6              |

## Calculate Statistic T:

$$T = \sum R_i Y_i = 2 + 4 + 5 + 7 + 6 = 24$$

## Distribution of T

- Let  $Y = \sum_{i=1}^n Y_i$  (Number of positive signs).
- $T = \sum R_i Y_i$  is the sum of ranks corresponding to positive values of  $X_i - \theta_0$ .
- Let  $r_1 < \dots < r_y$  be the ranks corresponding to positive differences vector of signs  $(r_1, \dots, r_n)$ .
- $Y \sim \text{Bin}(n, 0.5)$
- Under  $H_0$ , the vector of signs  $(Y_1, \dots, Y_n)$  can take any of the  $2^n$  values (e.g., 00...0 to 11...1) with equal probability  $\frac{1}{2^n}$ .

## Exact Null Distribution

Each configuration of signs gives a value for  $T$ . We count how many configurations sum to  $K$  to find  $P(T = K)$ .

**Example:**  $n = 7$ , Find  $P(T = 9)$  We need to find subsets of ranks  $\{1, 2, 3, 4, 5, 6, 7\}$  that sum to 9.

- Length 2: (2, 7), (3, 6), (4, 5)
- Length 3: (1, 2, 6), (1, 3, 5), (2, 3, 4)

Total combinations = 6.

$$P(T = 9) = \frac{6}{2^7}$$

**Other Probabilities ( $n = 7$ ):**

- $T = 0$ : Empty set (all signs 0).  $P(T = 0) = \frac{1}{2^n}$ .
- $T = 1$ : Set  $\{1\}$ .  $P(T = 1) = \frac{1}{2^n}$ .
- $T = 2$ : Set  $\{2\}$ .  $P(T = 2) = \frac{1}{2^n}$ .
- $T = 3$ : Sets  $\{3\}, \{1, 2\}$ .  $P(T = 3) = \frac{2}{2^n}$ .

The density is to evaluate the test statistic and find the p-value.

## Comparison of Tests

1. **t-test:** Assumes  $F \equiv N$  (Normal).
2. **Sign test:** Assumes  $F$  is continuous.
3. **Signed Rank test:** Assumes  $F$  is symmetric and continuous.

**Performance:**

- If the assumption of symmetry holds, the **Signed Rank test is more powerful** than the Sign test.
- If the assumption of symmetry does **not** hold, the p-value you get from the Signed Rank test will be **wrong**.

How do the tests compare

$$\text{one sample test for centre} \left\{ \begin{array}{ll} \text{t-test} & F \equiv N \\ \text{Sign test} & F \text{ is continuous} \\ \text{Signed Rank test} & F \text{ is symmetric \& continuous.} \end{array} \right. \text{ same level}$$

## Large Sample Approximation

**Distribution Parameters for T:**

- Mean:  $E[T] = \frac{n(n+1)}{4}$
- Variance:  $\text{Var}(T) = \frac{n(n+1)(2n+1)}{24}$

**Standardized Statistic:**

$$T^* = \frac{T - E[T]}{\sqrt{\text{Var}(T)}} \approx N(0, 1)$$

**Derivation:** Let  $V_i = R_i Y_i$ . Here ranks are fixed numbers  $1, \dots, n$  but  $Y_i$  is random. However, we can view this as:

$$T = \sum_{i=1}^n V_i$$

Where  $V_i$  takes the value  $i$  (if sign is positive) with prob  $1/2$ , and  $0$  (if sign is negative) with prob  $1/2$ .

### Abstract

We define  $V_i$  as the contribution of the  $i$ -th rank to the total sum  $T$ . Instead of summing randomized ranks, we look at the fixed rank value  $i$  (e.g., rank 1, rank 2, rank 5) and ask:

Will this rank be added to our total? If the sign is positive: We add the rank value  $i$  to our total. (Value =  $i$ ).

If the sign is negative: We do not add anything. (Value =  $0$ ).

Since the sign is a "coin flip" (50/50 chance) under  $H_0$ :  $V_i$  takes the value  $i$  with probability  $1/2$ .  $V_i$  takes the value  $0$  with probability  $1/2$

- $E[V_i] = i \cdot \frac{1}{2} + 0 \cdot \frac{1}{2} = \frac{i}{2}$
- $E[T] = \sum_{i=1}^n \frac{i}{2} = \frac{1}{2} \frac{n(n+1)}{2} = \frac{n(n+1)}{4}$
- $\text{Var}(V_i) = E[V_i^2] - (E[V_i])^2 = \frac{i^2}{2} - \frac{i^2}{4} = \frac{i^2}{4}$
- $\text{Var}(T) = \sum_{i=1}^n \frac{i^2}{4} = \frac{1}{4} \sum i^2 = \frac{1}{4} \frac{n(n+1)(2n+1)}{6} = \frac{n(n+1)(2n+1)}{24}$

**Note:**

- $V_i$  are independent as  $Y_i$  are independent.
- This approximation is useful when  $n$  is moderately large.
- Computation of  $2^n$  possibilities is very time consuming for large  $n$ .

## Handling Ties

What happens if data contains ties?

**Case 1: Zeros**

- If  $X_i - \theta_0 = 0$  for same  $i$ .
- **Method:** Drop those observations and reduce  $n$ .

**Case 2: Equal Absolute Differences**

- If  $|X_i - \theta_0| = |X_j - \theta_0|$  for  $i \neq j$
- **Method:** Give them the **Average Rank**.

**Example:** Values to rank: 5, 6, 5, 6, 7 1. Sort them: 5, 5, 6, 6, 7 2. Initial positions: 1, 2, 3, 4, 5 3. Average ties:

- The two 5s are at positions 1 and 2. Average rank =  $\frac{1+2}{2} = 1.5$ .
- The two 6s are at positions 3 and 4. Average rank =  $\frac{3+4}{2} = 3.5$ .
- The 7 is at position 5. Rank = 5.

4. Final Ranks: 1.5, 1.5, 3.5, 3.5, 5.

**Note:** When ties are present and average ranks are used, the p-value derived is **approximate**, not exact.

Week\_I

```
#This generates all possible subsets of size 2 when $n=4$.
combn(c(1:4),2)
```

```
#Writing own function for p-value of signed rank test
```

```
signrank<-function(n,T){
  x<-c(1:n)
  k<-0
  for(i in 1:n){
    y<-combn(x,i)
    l<-dim(y)[2]
    for(j in 1:l){
      k<-cbind(k,sum(x[y[,j]]))
    }
    pval<-length(which(k>=T))/length(k)
    return(pval)
  }
}
```

```
table(k)
```

```
#Signed rank test for the example done in class
```

```
a<-c(73, 82, 87, 68, 106, 60,97)
```

```
b<-a-70
```

```
r<-rank(abs(b))
```

```
s<-ifelse(b>0,1,0)
```

```
T<-sum(r*s)
```

```
signrank(7,T)
```

```
# Directly using built-in function for signed rank test
```

```
library(stats)
```

```
wilcox.test(a,mu=70,alternative="greater")
```

Week-II

```
# Wilcoxon rank sum test
```

```
# Example: Hollander & Wolfe (1973)
```

```
# Permeability constants of the human chorioamnion (a placental membrane) at
```

```
# term (x)
```

```
# and between 12 to 26 weeks gestational age (y). The alternative of interest
```

```
# is greater permeability of the human chorioamnion for the term pregnancy.
```

```
x <- c(0.80, 0.83, 1.89, 1.04, 1.45, 1.38, 1.91, 1.64, 0.73, 1.46)
```

```
y <- c(1.15, 0.88, 0.90, 0.74, 1.21)
```

```
z<-sort(c(x,y))
```

```
T<-sum(which(z %in% x))-10*11/2
```

```
T
```

```

ranksum<-function(n,m,T){
  T=T+m*(m+1)/2
  x<-c(1:n)
  l<-choose(n,m)
  k<-rep(0,l)
  y<-combn(x,m)
  for(j in 1:l){
    k[j]<-sum(x[y[,j]])}
    pval<-2*length(which(k<T))/choose(n,m)
    pval<-ifelse(pval>1, 2-pval, pval)
  return(pval)
}
ranksum(15,10,35)

wilcox.test( c(0.80, 0.83, 1.89, 1.04, 1.45, 1.38, 1.91, 1.64, 0.73, 1.46),
c(1.15, 0.88, 0.90, 0.74, 1.21))

# Permutation test using t-statistic
T<-t.test(x,y)$statistic
t.pval<-function(a,b,T){
  c<-cbind(a,b)
  m<-length(b)
  n<-length(c)
  x<-c(1:n)
  l<-choose(n,m)
  t<-rep(0,l)
  y<-combn(x,m)
  for(j in 1:l){
    t[j]<-t.test(c[y[,j]],c[-y[,j]])$statistic}
  pval<-2*length(which(t>=T))/l
  pval<-ifelse(pval>1, 2-pval, pval)
  return(pval)
}
t.pval(x,y,T)

```